Increasing Accuracy of Distant Supervision in Relation Extraction

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Proposed roles in the project:

- **Tianchen Xie** – Theoretical Design & Evaluation
  Focused on selecting models, developing the research methodology, and ensuring the project is grounded in solid theoretical principles.
- **Lewis Zhang** – Programming & System Development
  Responsible for implementing algorithms, coding experiments, and managing software infrastructure.
- **Jonathan Wu** – Evaluation & Analysis
  In charge of designing evaluation metrics, preparing data splits (training/dev/test), and conducting error analysis to validate system performance.
- **All three** – All three team members will contribute equally to writing, editing, and preparing the proposal and final report. We will collaborate to ensure clarity, coherence, and academic writing quality.

Problem statement

Relation extraction (RE) — identifying the relationship between two entities mentioned in a text — is vital for knowledge graph construction, question answering, and many downstream NLP applications. A widely used approach is distant supervision, where a knowledge base is used to automatically label sentences: if a known entity-pair appears in a sentence, that sentence is assumed to express the relation. While scalable, the method suffers from severe label noise: many sentences labelled positive do not actually express the target relation, leading to degraded performance in supervised models.

Our project investigates the question: Can we reduce the effect of noise from distant supervision using an attention mechanism? The approach we are considering is filtering sentences based on their trustworthiness. We aim to build a filter block based on attention to exclude noise from the raw data generated by distant supervision. We will first use a raw dataset to train our baseline model to see its basic performance. Afterwards, we will use the filter block to get a filtered dataset to train the RE model again. We will also use a small manually-labeled

dataset to train the model and use its performance as a theoretical upper bound for our model. Finally, we will compare the model's precision, recall, and F1 score on our filtered dataset with the one on the raw and manually labeled dataset to evaluate how our filter block works.

Evaluation plan

Task: Given a manually labeled test set of sentences grouped by entity pair ("bags"), classify each pair with the correct relation label (or no-relation).

Procedure: First, we will use a raw dataset to train our baseline model and keep track of its precision, recall, and F1 score. Then we will apply our filter to the raw dataset and get a "clean" dataset. We will use the filtered dataset to train the baseline model again and test it on our test set. We will also use a small manually-labeled dataset to train the model and use its performance as a theoretical upper bound for our model. Finally, we will compare the model's performance on the filtered dataset with the one on the original and manually labeled dataset to reflect our filter block's performance.

Datasets: We will obtain a publicly available distant‑supervision RE corpus (e.g., FB-NYT or NYT10). We will hand-annotate a small subset of the training data (e.g., ~100 bags) to serve as our test set.

Metrics:

Relation extraction: Precision, Recall, F-1 score on the test set.

Filtration analysis: average number of sentences per bag before and after filtering; proportion of removed positive sentences (ideally low).

Efficiency metrics: training time, model size, average sentences per bag, and overall throughput.

Error Analysis: We will perform error analysis to understand why the model succeeds or fails on specific examples. By inspecting sentence bags and their predicted relations, we aim to identify key error sources such as:

- Unreasonable threshold value, which may lead to too many or little sentences being filtered.
- Attention misallocation, where the model focuses on irrelevant sentences or tokens.
- Ambiguous expressions that require deeper contextual or syntactic understanding.

- Data bias or imbalance that skews relation predictions.

For each error type, we will review example sentences and discuss whether improvements—like tuning attention thresholds, refining features, or balancing data—could reduce these errors and enhance model reliability.

Former Research Papers

ARNOR: Attention Regularization based Noise Reduction for Distant Supervision Relation Classification (PDF - ACL Anthology)

This paper proposes "ARNOR", a framework that (1) uses attention mechanisms to find which sentence in each bag likely expresses the relation, and (2) introduces the attention regularization mechanism which encourages the model to focus on interpretable relations. Afterwards, the model selects trustworthy instances that can be clearly explained and re-trains the model using these selected instances. Their experiments on the NYT dataset show significant improvement over baseline distant-supervised RE.

Core idea and connection: The idea of selecting trustworthy training instances via attention is directly aligned with our thoughts. Our project will develop a similar filter block based on the attention mechanism to rescue noise generated by distance supervision to improve RE performances.

Potential extension: The use of attention regularization and trustworthy instance selection alone does not explore the potential use of manual annotations of which sentence in each bag best expresses the relation. Our project will also use a small number of manually-annotated relation entities to train the RC model as an upper-bound for us to reference and compare the performances of the models.

Neural Relation Extraction with Selective Attention over Instances (PDF - ACL Anthology)

This 2016 paper introduces a neural architecture for distantly supervised relation extraction that effectively handles noisy training data. Traditional distant supervision assumes that all sentences containing the same entity pair express the same relation, which leads to a large proportion of incorrect labeling. To address this, the authors propose a selective attention mechanism over instances, allowing the model to automatically assign different weights to sentences depending on their relevance to the

target relation. Each sentence is first encoded using a Convolutional Neural Network, and the attention mechanism aggregates these sentence representations into a single, noise-tolerant bag-level representation. Experiments on the NYT–Freebase dataset show that the proposed PCNN+ATT model achieves state-of-the-art performance, significantly outperforming previous multi-instance learning and feature-based approaches.

Core idea and Connection: The model, using an attention mechanism, learns to focus selectively on informative sentences among multiple instances for the same entity pair, rather than treating all of them equally. This strategy effectively reduces the impact of noisy supervision while capturing minor clues which average or max pooling cannot. Our project shares a similar idea of automatically adjusting the weight of the raw dataset (whether to filter the sentence out or not) and could potentially share a similar model architecture to reach desirable results.

Possible Improvements: While the paper's attention mechanism elegantly addresses noise at the sentence level, it still operates on sentence-level granularity and relies solely on CNN-based encoders. Potential improvements include incorporating contextualized language models (e.g., BERT, RoBERTa) to better capture cross-sentence dependencies or introducing multi-head attention to model finer-grained token-level relevance. For projects similar to ours, combining selective attention with modern transformer-based encoders and multi-instance reasoning could yield substantial performance gains and interpretability improvements.

Combining Distant and Direct Supervision for Neural Relation Extraction (PDF – ACL Anthology)

This paper reduces noise in distant supervision by combining distant and limited direct supervision in a multitask framework. Sentence-level labels guide attention to identify reliable sentences, improving relation detection and achieving a 4.4% AUC gain on FB-NYT. Their approach, supervising attention to filter noise, offers insights into effective attention designs, such as sigmoidal attention with max pooling, as alternatives to attention regularization or heuristic filtering.

Enhanced Distant Supervision with State-Change Information for Relation Extraction (PDF – ACL Anthology)

Enhanced Distant Supervision (EDS) is a framework that improves relation extraction by integrating time encoding information into distant supervision. Instead of assuming static relations, EDS filters training data based on whether a sentence's timestamp aligns with a relation's valid time window, reducing noisy labels. This method also enables state-change relation extraction. Their results show that EDS improves precision and provides a benchmark for temporally-aware RE. The alignment heuristic can serve as a preprocessing or weighting step in other pipelines and could be extended with explicit temporal reasoning or temporal knowledge graph representations.

Improving Relation Extraction with Knowledge-attention (PDF – ACL Anthology)

The paper introduces a knowledge-attention mechanism for neural relation extraction that integrates external knowledge-base information into attention, guiding focus toward relation-relevant tokens. This fully attention-based model (without CNNs or RNNs) achieves state-of-the-art results on TACRED by fusing prior knowledge with self-attention through various integration schemes. Knowledge-attention acts as a soft regularizer that reduces noise and enhances interpretability.

A Strategy for Solving the Problem

Implementation: We use a two-stage, modular pipeline built on the common bag-level attention approach for distant supervision.

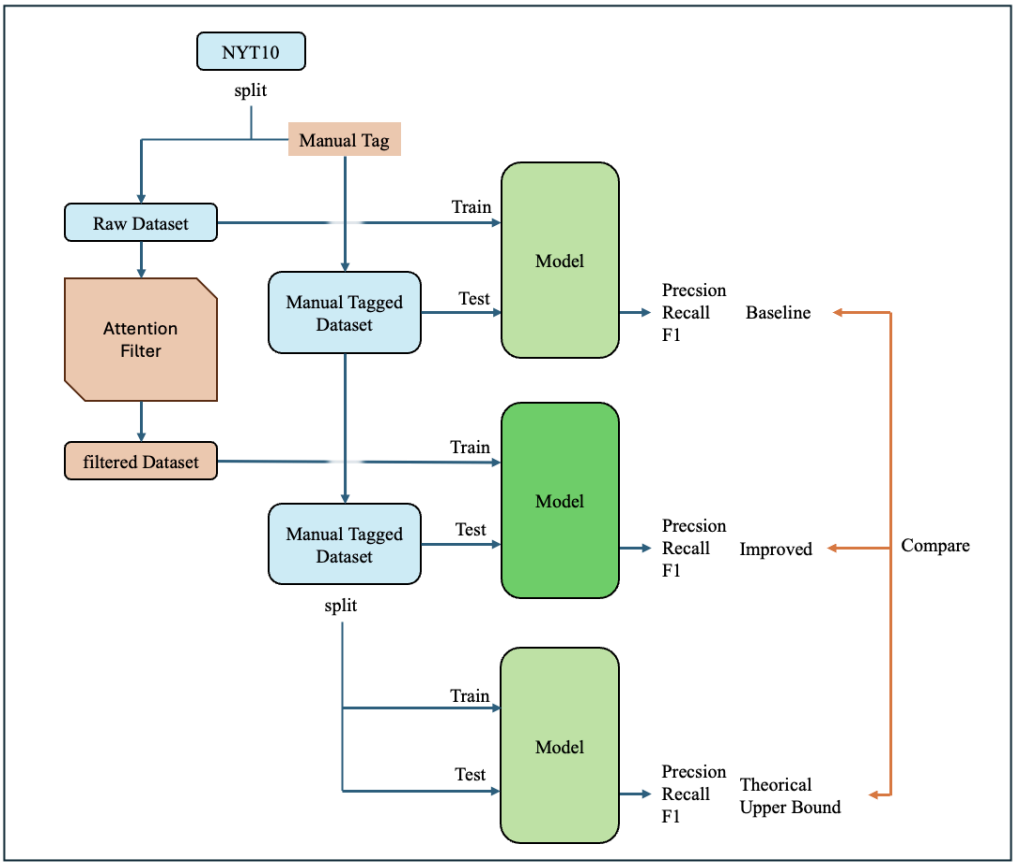
Stage 1: Use a BERT-based sentence encoder and train a bag-level classifier that assigns an attention weight to each sentence (higher weight = more useful for predicting the relation). Train this baseline on a small, controlled subset of a distant-supervision corpus (e.g., NYT10 downsampled to 5–10 relations and ≤10k bags) to obtain stable attention distributions.

Stage 2: Use the learned per-sentence attention to filter training data and retrain. We will choose from or compare between the three practical filtering rules and tune them on a development set: (A) absolute threshold: remove sentences whose attention is below a chosen value; (B) top-k: keep only the k highest-attention sentences per bag; (C) relative margin: keep the top sentence only when its attention substantially exceeds the next best. To avoid losing all evidence, we employ a conservative fallback that always preserves at

least one sentence per bag. After filtering, we retrain the relation model on the cleaned data and compare it to the baseline.

We will also collect a small hand-labeled set (about 100–200 bags) indicating which sentence truly expresses the relation and use it to directly supervise attention during training; This shall serve as a theoretical upper bound for our baseline model's performance. By comparing the model's performance on the filtered dataset with the raw and manually-labeled one, we can get a more comprehensive picture of our filter block's performance.

Elaboration if time permits: 1) Using multi-head sentence attention with sigmoidal gates to improve the attention weight of sentence vectors. 2) Self-training loop: after each round, re-filter using updated attention, retrain classifier.



A collaboration plan.

Besides the “Proposed roles in the project” at the beginning of this proposal, our project will follow a modular pipeline design so all members can work in parallel. The system is divided into three stages—baseline modeling, attention-based filtering, and evaluation—each with fixed input/output formats to ensure compatibility. To allow independent progress, we will use sample “gold” data that mimics expected outputs, so modules can be developed and tested before integration.

All code and data will be managed through a shared GitHub repository with standardized JSON/CSV formats and version control. Each component will include minimal tests to verify correctness, and integration will proceed in stages once individual parts are stable. Weekly meetings and group chat will coordinate interface updates and resolve issues quickly. This setup enables smooth collaboration and simultaneous development without workflow bottlenecks.

References

Beltagy, I., Lo, K., & Ammar, W. (2019). Combining distant and direct supervision for Neural Relation Extraction. *Proceedings of the 2019 Conference of the North*, 1858–1867. https://doi.org/10.18653/v1/n19-1184

Jia, W., Dai, D., Xiao, X., & Wu, H. (2019). Arnor: Attention regularization based noise reduction for Distant Supervision Relation Classification. *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, 1399–1408. https://doi.org/10.18653/v1/p19-1135

Li, P., Mao, K., Yang, X., & Li, Q. (2019). Improving relation extraction with knowledge-attention. *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, 229–239. https://doi.org/10.18653/v1/d19-1022

Lin, Y., Shen, S., Liu, Z., Luan, H., & Sun, M. (2016). Neural relation extraction with selective attention over instances. Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers). https://doi.org/10.18653/v1/p16-1200

Shah, J., Zhang, D., Brody, S., & McCallum, A. (2022). Enhanced Distant Supervision with State-Change Information for Relation Extraction. *Proceedings of the Thirteenth Language* Resources and Evaluation Conference **.** https://aclanthology.org/2022.lrec-1.597.pdf